Lecture 26

Appendix B

Complex Numbers

Note: In lecture 23, we noted that the characteristic polynomial of an $n \times n$ matrix has at most n real roots (counting multiplicity), and so an $n \times n$ matrix has at most n real eigenvalues. If we were to extend our scalars to the complex numbers (and not just consider the real numbers), then our characteristic polynomial would indeed have n roots in the complex numbers (counting multiplicity).

Definition: A complex number is a number written in the form $z = a + bi$, where a and b are real numbers and $i = \sqrt{-1}$. The real part of z is a, denoted by $\operatorname{Re} z$. The imaginary part of z is b, denoted by $\operatorname{Im} z$.

Note: Two complex numbers are equal if and only if their real and imaginary parts are equal.

Note: A real number a is a complex number, i.e. $a = a + 0i$.

Definition: We denote the complex number system, i.e. the set of all complex numbers, by $\mathbb{C}$.

We can naturally define addition and multiplication of complex numbers by

$$(a + bi) + (c + di) = (a + c) + (b + d)i,$$
$$(a + bi)(c + di) = (ac - bd) + (ad + bc)i.$$

Definition: The conjugate of $z = a + bi$ is the complex number $\overline{z}$, namely "z bar," defined by $\overline{z} = a - bi$.

Remark: If $z = a + bi$, then

$$z\overline{z} = (a + bi)(a - bi) = a^2 - abi + abi - b^2 i^2 = a^2 + b^2 \in \mathbb{R}.$$

Definition: The absolute value (or modulus) of $z = a + bi$ is the real number $|z|$ defined by

$$|z| = \sqrt{z\overline{z}} = \sqrt{a^2 + b^2}.$$

Useful Properties: Suppose w and z are complex numbers.

1. If z is a real number, then $z = a + 0i$ for some real number a, and so $|z| = \sqrt{a^2} = |a|$. In other words, the modulus, in this case, is the same as the absolute value.

2. $\overline{z} = z$ if and only if z is a real number.

3. $\overline{w + z} = \overline{w} + \overline{z}$.

4. $\overline{wz} = \overline{w}\,\overline{z}$. In particular, $\overline{rz} = r\overline{z}$ if r is a real number.

5. $z\overline{z} = |z|^2 \geq 0$.

6. $|wz| = |w||z|$.

7. $|w + z| \leq |w| + |z|$.

Note: If $z \neq 0$, then $|z| > 0$, and so z has a multiplicative inverse, denoted by $\frac{1}{z}$ or z^{-1}, where

$$z^{-1} = \frac{\overline{z}}{|z|^2}.$$

So, if $z = a + bi$, then $z^{-1} = \frac{a-bi}{a^2+b^2}$.

Example: Write

$$\frac{1+3i}{2+4i}$$

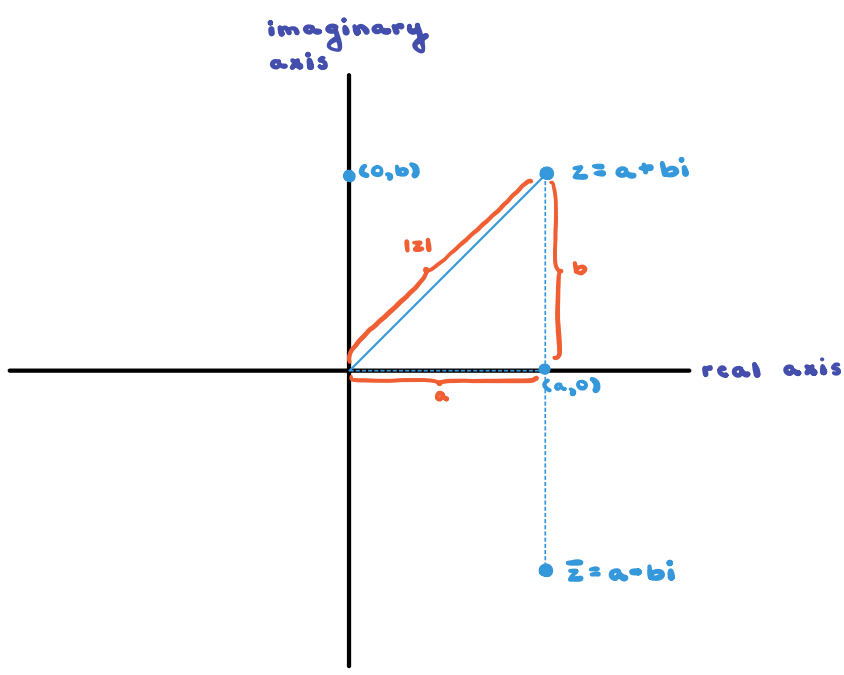
as a complex number of the form $a + bi$.

Solution: We see that

$$\frac{1+3i}{2+4i} = \frac{1+3i}{2+4i} \cdot \frac{2-4i}{2-4i} = \frac{2-4i+6i+12}{2^2+4^2} = \frac{14+2i}{20} = \frac{7}{10} + i\frac{1}{10}.$$

Geometry of $\mathbb{C}$

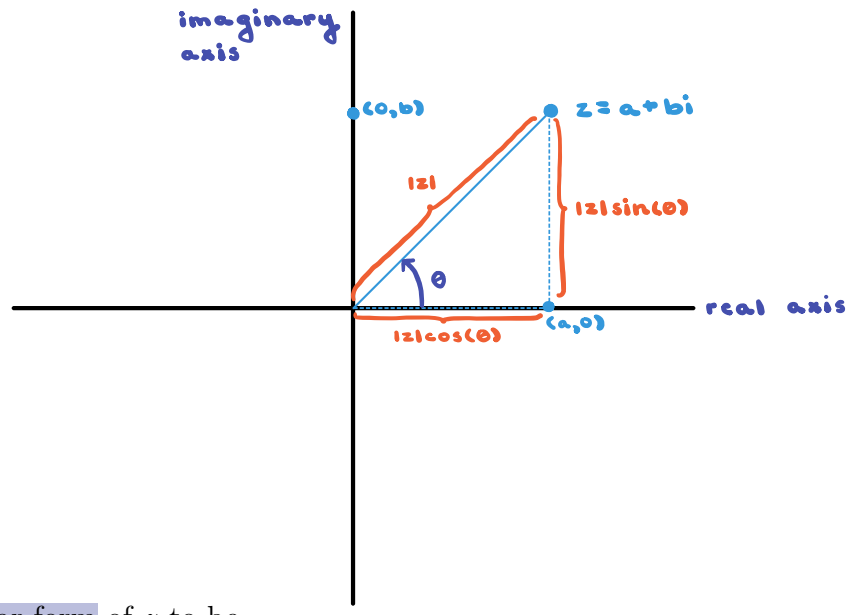
Each complex number $z = a + bi$ corresponds to a point (a, b) in $\mathbb{R}^2$. We can think of $\mathbb{C}$ as a Cartesian plane, where the horizontal axis is the real axis, and the vertical axis is the imaginary axis.



Polar Coordinates

Given a nonzero complex number $z = a + bi$, we let θ be the angle between the positive real axis and the point (a, b). The angle θ is the argument of z, i.e. $\theta = \arg z$. Using trigonometric identities, we have that

$$\frac{a}{|z|} = \cos(\theta) \implies a = |z|\cos(\theta),$$
$$\frac{b}{|z|} = \sin(\theta) \implies b = |z|\sin(\theta).$$

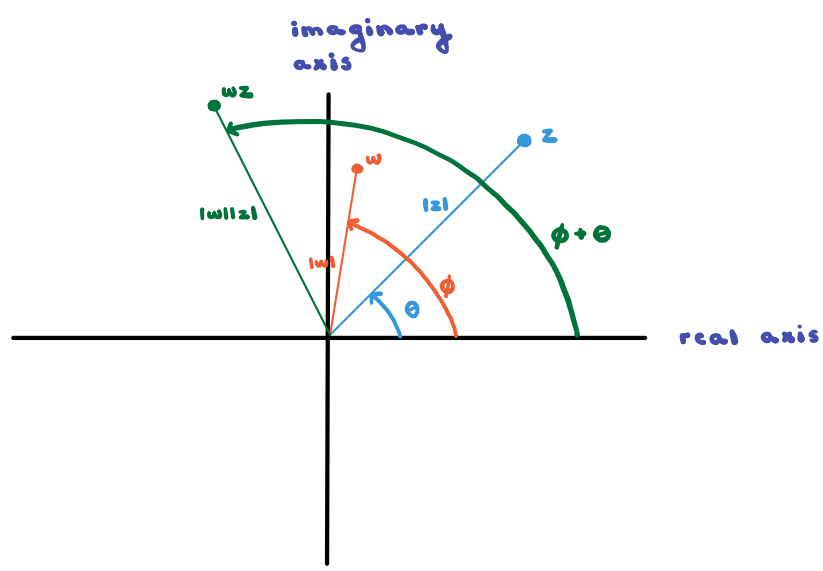


So, we define the polar form of z to be

$$z = a + bi = |z|\cos(\theta) + |z|\sin(\theta)i = |z|(\cos(\theta) + i\sin(\theta)).$$

Note: To convert a complex number $z = a + bi$ into polar form, we will need $|z| = \sqrt{a^2 + b^2}$ and $\theta = \tan^{-1}\left(\frac{b}{a}\right)$. Also, using the "product to sum formulas" (from trigonometry), if we have another complex number $w = |w|(\cos(\phi) + i\sin(\phi)) \neq 0$, then

$$wz = |w||z|(\cos(\phi + \theta) + i\sin(\phi + \theta)).$$



Theorem (De Moivre): For any positive integer k, if $z = r(\cos(\theta) + i\sin(\theta))$, then

$$z^k = r^k(\cos(k\theta) + i\sin(k\theta)).$$

Theorem: $e^{i\theta} = \cos(\theta) + i\sin(\theta)$

Proof: If we apply the Maclaurin series formula to $e^{i\theta}$, we get that

$$\begin{aligned}
e^{i\theta} &= \sum_{n=0}^{\infty} \frac{(i\theta)^n}{n!} \\
&= 1 + i\theta - \frac{\theta^2}{2!} - i\frac{\theta^3}{3!} + \frac{\theta^4}{4!} + i\frac{\theta^5}{5!} + \cdots \\
&= \left(1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} + \cdots\right) + i\left(\theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} + \cdots\right) \\
&= \cos(\theta) + i\sin(\theta).
\end{aligned}$$

It follows that, if $\arg z = \theta$, then

$$z = |z|(\cos(\theta) + i\sin(\theta)) = |z|e^{i\theta}.$$

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Example: Find all real and complex solutions of $z^7 = 11^7$.

Solution: Let $z = re^{i\theta}$. Then,

$$(re^{i\theta})^7 = 11^7 \implies r^7e^{i7\theta} = 11^7 = 11^7e^{i0}.$$

Using our theorem above, we have that

$$r^7(\cos(7\theta) + i\sin(7\theta)) = 11^7(\cos(0) + i\sin(0)) = 11^7(1 + i0),$$

implying that $r = 11$, $\cos(7\theta) = 1$, and $\sin(7\theta) = 0$. We know that $\cos(7\theta) = 1$ and $\sin(7\theta) = 0$ when $7\theta = 0 + 2\pi k$ for $k \in \mathbb{Z}$, i.e. $\theta = \frac{2\pi}{7}k$ for $k \in \mathbb{Z}$. So, it must be that

$$z = 11e^{i\frac{2\pi}{7}k},$$

where $k = 0, 1, 2, \ldots, 6$. Notice that after $k = 6$, the solutions begin to repeat. We only have 7 unique solutions.